Designing the Database

Database Design and Administration (Satzinger et al., 2016)

- A **database (DB)** is an integrated collection of stored data that is centrally managed and controlled. A database typically stores information about dozens or hundreds of classes or entities. A database is managed and controlled by a database management system
- A **database management system (DBMS)** is a system software component that is generally purchased and installed separately from other system software components (e.g., operating systems).
- ***Figure 1*** illustrates the components of a typical database. Usually, when we refer to a database, it means both the database data, shown on the left and the DBMS, shown in the middle.

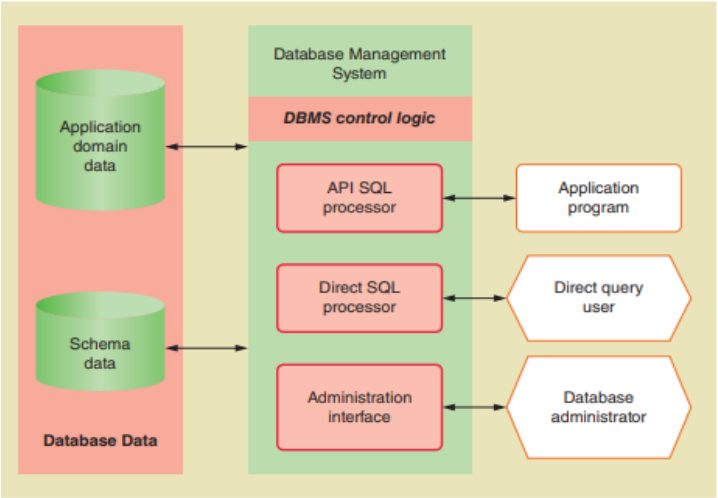

Figure 1. Database and DBMS Components with Interacting Actors

- The database data consists of two related information stores: the **application domain data**, which is the actual data for the problem domain, and the **schema**, which contains descriptive information about the application domain data.
- The schema data is required because it describes the structure and rules for accessing the application domain data. It is often referred to as metadata or data about the data. It includes the following:
  - Organization of individual stored data items into higher-level groups, such as tables
  - Associations among tables (e.g., pointers from customer objects to related sale objects)
  - Details of individual data items, including types, lengths, locations, and indexing of data items
  - Access and content controls, including allowable values for specific data items, value dependencies among multiple data items, and lists of users allowed to read or update data items.
- **Database design** is defined as a collection of steps that help with designing, creating, implementing, and maintaining a business's data management systems
- It refers to the activities that focus on the design of the database structure that will be used to store and manage end-user data
- The main purpose of designing a database is to produce physical and logical models of designs for the proposed database system.

Database Design Team (Satzinger et al., 2016)

- Larger organizations typically have staff members dedicated to maintaining the operational databases that underlie an organization's information systems. Although the project team has primary responsibility for defining the domain model and designing the corresponding database for a new system, permanent database staff usually assist the project team in database design
- It ensures consistency between database requirements for new and existing systems and helps to avoid problems that might disrupt existing systems when a new system is deployed
- Two key positions on the permanent database staff are the data administrator and the database administrator.
- A **data administrator (DA)** has responsibility for the structure and integrity of the data itself. Specifically, the DA manages

important aspects of data definition and database design, including the following:

- **Data standards.** Naming standards, definition standards, data typing standards, and value edits
- **Data use.** Ownership of data, accessibility of data, and confidentiality
- **Data quality.** Validation rules, completeness, currency, consistency, and relevancy

- A **database administrator (DBA)** maintains the database after it has been deployed and manages the safety and operation of the database
- A database administrator's job is to ensure that the database, both the data and the DBMS, is configured correctly for the organization's architecture and performs effectively and efficiently. The DBA's responsibilities include the following:
  - Managing a multiple DBMS environment
  - Protecting the data and database, including user authentication and attack prevention
  - Monitoring and maintaining high levels of performance
  - Backing up the database and defining recovery procedures.
- ***Figure 2*** illustrates the different focus and responsibilities of the data administrator and the database administrator.

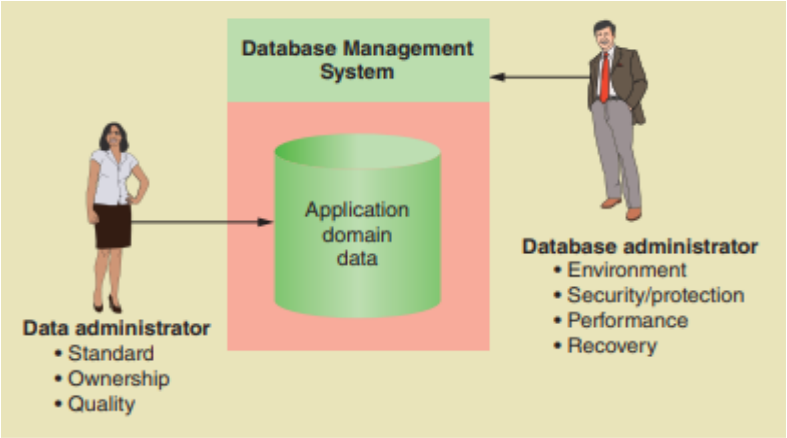

Figure 2. Data and database administrators' responsibilities

Relational Database

- A **relational database management system (RDBMS)** is a DBMS that organizes stored data into structures called tables or relations
- A **table** (**relation**) is a two-dimensional structure composed of rows and columns
- Each table row (**tuple**) represents data.
- Each table column represents an **attribute**, and each column has a distinct name.
- Each intersection of a row and column represents a single data value.

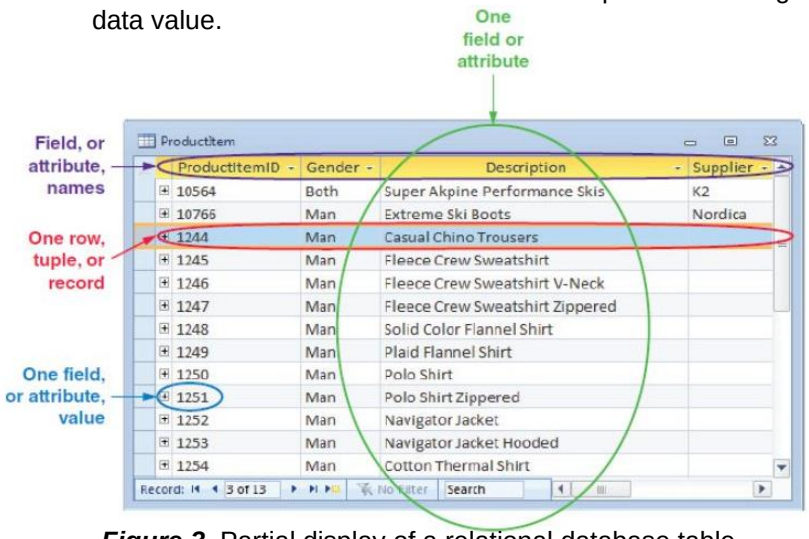

Figure 3. Partial display of a relational database table

- A **key** is an attribute or set of attributes; the values of which occur only once in all the rows of the table.
- A key is an attribute or group of attributes that determine the values of other attributes. For example, an invoice number identifies all the invoice attributes, such as the invoice date and the customer name.

- The types of keys are the following:

Key Type	Description
Superkey	An attribute or combination of attributes that uniquely identifies any row in the table

Example	`STU_NUM` and any combination with `STU_NUM`
	Ex. `(STU_NUM, STU_LNAME)`
	`(STU_NUM, STU_FNAME, STU_SECT)`

Candidate Key	A superkey without any unnecessary attributes
Example	`STU_NUM`
Primary Key	A candidate key selected to uniquely identify all other attribute values in any given row; cannot contain null entries.
Example	`STU_NUM`
Foreign Key	An attribute or combination of attributes in one table whose values must either match the primary key in another table or be null.
Example	`STU_SECT` can be a foreign key if it is used as a primary key of another table.
Secondary Key	An attribute or combination of attributes used strictly for data retrieval purposes.
Example	`(STU_LNAME, STU_FNAME, STU_MI)`

Protecting the Database (Satzinger et al., 2016)

- **Transaction logging** is a technique by which any update to the database is logged with such audit information as user ID, date, time, input data, and type of update.
- Transaction logging provides a record of database changes that are stored in a separate location and can be checked independently of the database itself.
- Transaction logging achieves two objectives.
  - First, it helps discourage fraudulent transactions or malicious database changes.
  - Second, a logging system provides a recovery mechanism for erroneous transactions.
- Large active databases often have tens of thousands of users every minute who need to read and update the data. Not only do databases need to be extremely fast, but they also need to be designed so that multiple users do not damage each other's data inadvertently when they are accessing or updating the same information.
- To achieve this, techniques that implement two concepts are used:
  - A **transaction** is a piece of work that has several steps, including several reads and writes to the database, that must all be completed to be valid. An example might be buying movie tickets online. The transaction will require the following steps:
    1. Read the database for available seats and display it.
    2. Update the database when the user chooses seats.
    3. Start a timer.
    4. Accept user purchase information (credit card number).
    5. Verify credit and send a charge to the user's credit card.
    6. Post the user's payment to the company ledger.
    7. Permanently update the seat database that the chosen seats are now taken.
    8. Update the user information with a reservation for the specific seats.
  - All these steps must be completed for the "purchase ticket transaction" to be complete. If the user stops, if the credit card charge is refused, or if the timer runs out, then all the prior changes must be undone, and the database returned to its original condition.
  - The second important concept that is widely used to protect against lost updates is to apply a **database lock**. A database lock is a technique where a portion of the database is locked by one user so that no other users (or programs) can use that portion of the database until the first user is finished and releases the lock.
  - There are two types of locks that are used, a shared lock, also called a read lock, and an exclusive lock, also called a write lock.

- If the user application only needs to read the data, with no need to update it, it can issue a **shared lock**. A shared lock allows other users to also read the same data but not to update it.
- If the user application needs to read and then rewrite the data, then an **exclusive lock** is issued. No other user can even read the data that has an exclusive lock.

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